

Dane Williamson

PhD. Candidate, University of Virginia

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Research Interest

My research is in robust, interpretable machine learning, with a focus on representation learning for high-dimensional data. My work Leverages Deep Generative Models to uncover meaningful latent structures, with applications in various complex domains. With the goal of developing scalable, interpretable frameworks that enhance model robustness and bridge theoretical advances with real-world impact.

Education

AUG. 2020 – **Ph.D. (UVA) University of Virginia, Charlottesville, VA**, Major: Computer Science
CURRENT Advisor: Yangfeng Ji

May 2020 **B.Sc. (VSU) Virginia State University, Petersburg, VA**, Major: Computer Science
Capstone Project: Cross Platform Mobile Application
Advisor: Prof. Ahmed Mohammed

Experience

- May – Aug. **MLE Intern, Stripe, NYC, NY**
2026
- Developed a hybrid multi-task learning and transfer learning architecture for fraud detection across heterogeneous merchant populations, using population-specific prediction heads.
 - Designed real-time model-serving infrastructure to publish fraud scores through Kafka topics and route each transaction to the appropriate merchant-population head.
 - Identified class imbalance as the source of gradient starvation for an underserved merchant population and implemented a positive-class reweighting strategy that improved performance across populations, especially for the affected group.
 - Proposed a data valuation and active learning pipeline to quantify the usefulness of training examples and improve sample selection for human annotation and model retraining.
- May – Aug. **Research Intern, Oracle Machine Learning Research Group, Burlington, MA**
2022
- Developed Named Entity Extraction System for Oracle Intelligent Document Recognition.
 - Implemented 2 Dimensional Positional Embeddings to leverage document layout information.
 - Introduced Special tokens to BERT tokenizer to prevent fragmentation of numerical tokens.
 - Improved F-1 Score on identifying 29/36 Entities over Conditional Random Fields (CRFs).
- May – Aug. **Research Intern, Oracle Machine Learning Research Group, Burlington, MA**
2021
- Pretrained BERT transformers using internal toolkit.
 - Designed and implemented Probes for knowledge and language understanding in BERT transformers.
 - Evaluated model performance on GLUE benchmarks using the Jiant toolkit.
- May – Aug. **Software Engineer Intern, Apple, Cupertino, CA**
2019
- Developed Python Desktop Application to simulate Host Stack of third party BLE Devices.
 - Developed Web-Scraper to populate MongoDB database with SIG Bluetooth Specifications.
 - Utilized internal Python-based framework to communicate with BLE controller via HCI commands.
 - Simulated BLE GATT Server using database containing specifications and HCI Commands.
- May – Aug. **Software Engineer Intern, IBM, Raleigh, NC**
2018
- Wrote automated end-to-end user-interface tests in Protractor Framework for Watson Content Hub.
 - Utilized JavaScript/TypeScript to conduct automated API testing in Staging and Production environments.
 - Configured Continuous Integration (CI) testing jobs with Jenkins pipeline builds for Docker and NPM registries.

Profile Summary

Selected Awards

- 2019 **Best Paper Award for the Human-Computer Interaction Thematic Area**, *HCI International 2019*, Orlando, FL, USA, *July 16-31, 2019*
Awarded for exceptional research contribution in the Human-Computer Interaction thematic area.

Grants, Scholarships & Fellowships

- 2016-2020 **EST-Univ STEM Scholarship**
2016-2020 **CIAA Scholarship**
2019 **Apple Engineering Scholarship**

Publications

Journal Papers

- 2026 **D. Williamson, Y. Ji, M. Dwyer**, *Contrastive Activation for Class-Sensitive Explanations*, *Machine Learning*
Accepted for publication by Springer Nature; preprint available at arXiv:2506.07327

In Submission

- 2025 **D. Williamson, Y. Ji, M. Dwyer**, *Optimizing Latent Dimension Allocation in Hierarchical VAEs: Balancing Attenuation and Information Retention for OOD Detection*
Preprint available at arXiv:2506.10089

Conference Papers

- 2025 **D. Williamson, Y. Ji, M. Dwyer**, *Syntactic Blind Spots: How Misalignment Leads to LLMs' Mathematical Errors*, *Proceedings of the 3rd Workshop on Mathematical Natural Language Processing (MathNLP 2025)*, *EMNLP 2025*, Suzhou, China
- 2019 **G. Cowart, D. Williamson, N. Farhat, and J.S. Lee**, *'Do Humans STILL Have a Monopoly on Creativity or Is Creativity Overrated?'*, *Human-Computer Interaction. Perspectives on Design – Thematic Area, HCI 2019*, 21st HCI International Conference, HCII 2019, Orlando, FL, USA
Presented at HCI 2019, July 26–31, 2019, and published in the Proceedings, Part I.

Conference Presentations

- 2026 **J. Shen, A. Castillo, D. Williamson, A. El Fathi, M. Breton**, *Enhancing Safety in Neural Network-Based Artificial Pancreas System via Variational Autoencoder Outlier Detection*, *Break-through T1D Clinical & Research Congress (CRC 2026)*
Abstract accepted for a Long Oral Presentation; preprint available at SSRN:6730495

Mentorship and Teaching

Mentorship

- May 2022 – **Andrew Kelly**, *Morgan State University*, Undergraduate Student
August 2022 Interpretability project on saliency analysis for Image Classifiers

Teaching Experience

- January 2022 – May 2024 **Graduate Teaching Assistant**, *School of Engineering and Applied Science, University of Virginia*
- CS 6316: Machine Learning, **Spring 2022**
 - CS 4774: Machine Learning, **Spring 2023**
 - CS 6501: Natural Language Processing, **Fall 2023**
 - CS 4710: AI, **Spring 2024**

- Aug. 2019 – **Undergraduate Teaching Assistant, Engineering and Computer Science Department, Virginia State University**
May 2020
- CSCI 120: Introduction to Programming, **Spring 2019**
 - CSCI 150: Programming I, **Spring 2019**
 - CSCI 151: Programming I Lab, **Spring 2019**
 - CSCI 250: Programming II, **Spring 2019**
 - CSCI 251: Programming II Lab, **Spring 2019**
 - CSCI 287: Data Structures, **Spring 2019**
 - CSCI 296: Web Programming, **Spring 2019**

Membership and Professional Service

Reviewer

- 2024 **EMNLP**, *Empirical Methods in Natural Language Processing*, Miami, Florida
2023 - **ACL ARR**, *Association for Computational Linguistics*, Rolling
Present

Member

- 2016 - **ACM**, *Association for Computing Machinery*
2016 - **NSBE**, *National Society of Black Engineers*